

TYUKALINSK, Russian Federation

WMO: 285860

Lat: 55.867N

Lon: 72.217E

Elev: 118

StdP: 99.92

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.7	-31.6	-37.5	0.1	-34.5	-34.5	0.1	-31.5	8.3	-10.6	7.6	-10.5	1.1	100	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	29.3	19.2	27.6	18.5	25.9	17.9	20.7	26.7	19.7	25.3	18.9	24.2	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	23.1	17.7	12.8	22.5	16.7	12.1	21.8	60.2	26.8	57.1	25.4	54.2	24.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	DB	-36.8	32.3	3.6	2.2	-39.5	33.8	-41.6	35.1	-43.6	36.3	-46.3	37.9	(4)
(5)			WB	-36.8	22.1	3.8	1.0	-39.5	22.8	-41.8	23.4	-43.9	24.0	-46.7	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-18.2	-15.6	-6.7	3.2	11.6	16.9	18.3	16.1	9.9	3.0	-6.6	-14.9
	DBStd	14.15	7.73	7.20	5.74	5.61	5.23	4.43	3.71	3.92	4.26	4.81	7.19	8.01
	HDD10.0	3921	875	718	518	213	45	4	0	2	52	221	499	773
	HDD18.3	6261	1133	951	777	455	216	77	48	92	255	476	749	1032
	CDD10.0	816	0	0	0	8	94	212	257	192	49	3	0	0
	CDD18.3	114	0	0	0	0	7	34	47	24	2	0	0	0
	CDH23.3	1096	0	0	0	3	112	344	405	209	23	0	0	0
	CDH26.7	260	0	0	0	0	26	92	94	45	3	0	0	0
Wind	WSAvg	3.0	2.8	2.8	3.4	3.5	3.5	2.9	2.7	2.7	2.8	3.1	3.3	3.1
Precipitation	PrecAvg	417	16	12	16	25	38	59	71	62	37	31	28	22
	PrecMax	612	28	24	49	48	82	105	213	128	75	61	55	39
	PrecMin	302	1	1	1	3	10	25	21	22	11	8	8	8
	PrecStd	78	8	5	11	13	20	21	42	26	17	16	12	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.4	1.2	6.4	22.6	29.5	31.3	31.6	30.0	26.6	17.3	6.5	1.6
		MCWB	-1.3	-0.4	3.3	12.4	15.9	18.7	20.7	19.7	16.0	10.4	4.9	0.8
	2%	DB	-2.3	-0.6	4.0	17.8	26.2	29.2	29.1	27.7	22.3	14.9	4.6	-0.1
		MCWB	-2.9	-1.7	1.6	10.2	15.2	19.1	19.8	19.0	13.9	9.4	3.4	-0.9
	5%	DB	-4.1	-2.5	2.3	14.9	23.8	27.1	27.2	25.7	19.5	12.3	3.1	-2.0
		MCWB	-4.6	-3.2	0.7	8.9	14.3	18.4	19.1	18.1	13.4	8.1	1.9	-2.9
	10%	DB	-6.5	-4.8	1.0	12.0	21.1	25.0	25.5	23.5	17.2	10.1	1.5	-4.0
		MCWB	-7.1	-5.6	-0.4	7.2	12.9	17.6	18.6	17.3	12.2	7.1	0.5	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	-0.2	4.0	13.2	17.7	21.9	22.0	21.2	17.1	11.9	5.1	0.9
		MCDB	-0.5	0.7	5.7	20.7	25.2	27.8	28.2	27.4	23.5	15.6	6.0	1.5
	2%	WB	-2.9	-1.6	2.0	10.8	16.1	20.2	20.9	20.0	15.3	10.2	3.7	-0.7
		MCDB	-2.4	-0.8	3.4	16.7	24.2	26.1	27.0	25.7	20.2	13.6	4.6	-0.2
	5%	WB	-4.6	-3.2	0.8	9.1	14.8	19.2	20.0	18.9	13.9	8.7	1.9	-2.8
		MCDB	-4.1	-2.4	2.1	14.3	22.3	25.2	25.5	24.0	18.6	11.6	2.9	-2.0
	10%	WB	-7.1	-5.6	-0.3	7.5	13.6	18.2	19.2	17.9	12.5	7.2	0.6	-4.7
		MCDB	-6.6	-4.8	1.0	11.8	20.2	23.9	24.3	22.4	16.7	10.0	1.4	-4.0
Mean Daily Temperature Range		MDBR	7.6	8.9	8.9	9.1	11.1	10.2	9.7	9.7	8.8	6.9	5.9	7.5
	5% DB	MCDBR	7.6	7.0	8.0	14.0	15.3	13.7	12.5	13.4	13.3	11.2	5.8	6.5
		MCWBR	7.5	6.5	6.4	8.1	6.8	5.9	5.1	6.0	7.0	6.8	5.1	6.4
	5% WB	MCDBR	7.5	6.9	7.4	13.1	14.0	12.0	11.0	11.4	11.8	9.6	5.5	6.5
		MCWBR	7.4	6.6	6.2	8.0	6.9	5.7	5.2	5.7	6.9	6.6	5.1	6.4
Clear-Sky Solar Irradiance	taub	0.278	0.312	0.368	0.412	0.404	0.407	0.429	0.430	0.371	0.354	0.315	0.283	
	taud	2.043	2.021	1.939	2.085	2.233	2.295	2.229	2.193	2.323	2.222	2.090	1.990	
	Ebn at Noon	595	722	765	798	836	838	807	775	777	676	537	471	
	Edh at Noon	65	98	138	139	128	122	128	124	95	81	62	54	
All-Sky Solar Radiation	RadAvg	0.87	1.93	3.48	4.60	5.45	5.85	5.45	4.30	2.93	1.50	0.73	0.55	
	RadStd	0.07	0.10	0.24	0.35	0.44	0.42	0.40	0.38	0.28	0.18	0.12	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page